This plan will strengthen our national diagnostic capability and capacity for terrestrial animals. It shows how we will maintain and improve our animal health laboratory system.

We developed the 2021-2026 plan with:

- animal health laboratories
- governments
- universities
- industry stakeholders.

The plan is driven by the principle that biosecurity is a shared responsibility. It works within the all-inclusive framework of Animalplan 2022-2027.

Objectives

Objectives of the current plan build on existing strengths and address identified gaps. It aims to maintain and improve our animal health diagnostic capability and capacity.

The activities against 5 key objectives of the plan include:

- implementing diagnostic high-throughput sequencing
- increasing laboratory surge capacity
- increasing capacity to detect emergency animal diseases
- point-of-care test validation
- standardised antimicrobial resistance tests.

National Animal Health Diagnostics Business Plan 2021-2026

The NAHDBP represents Australia's ongoing commitment to maintaining and improving diagnostic capability and capacity for terrestrial animals.

If you have difficulty accessing these files, contact us for help.

Projects

The plan guides the delivery of nationally coordinated activities.

Below is a high-level overview of some of the projects we support, and contribute to, under the plan.

The Animal Health Committee provides guidance and oversight.

Find more information about the projects.

Current Projects

Northern Australian biosecurity sequencing

- Establish high-throughput sequencing capability and capacity in Northern Australia.

National capacity building for diagnostic high-throughput sequencing

- Develop routine diagnostic high-throughput sequencing and bioinformatics capacity in animal health laboratories.

Completed Projects

Lumpy skin disease and African horse sickness sequencing workflow

- Develop quality-assured whole genome sequencing protocols for use in an outbreak.

Quality assured sequencing for infectious agent discovery

- Establish standardised high-throughput sequencing workflows for infectious agent discovery.

Australian Biosecurity Genomic Database

- Develop a database for notifiable terrestrial and aquatic animal diseases in Australia.

Current projects

Sample Tracking and Reporting System (STARS) version 2 deployment

- Improve information sharing between animal health laboratories.

Completed projects

Sample Tracking and Reporting System (STARS) enhancement

- Improve information sharing between animal health laboratories.

Exercise Waterhole: national laboratory simulation exercise

- Test and evaluate Australia’s national laboratory preparedness for an outbreak. See Exercise Waterhole.

National identification system for animal diagnostics

- Trial pre-barcoded sample tubes to increase laboratory efficiency in an outbreak.

National laboratory stockpile for emergency animal diseases responses

- Scope a centralised national stockpile for animal health laboratories.

Current Projects

Enhancing diagnostic capability for glanders

- Establish an accredited diagnostic process for glanders.

Equine piroplasmosis diagnostic capability development

- Develop molecular detection tests to screen horses entering or exiting Australia.

Completed Projects

Lumpy skin disease testing in cattle and buffalo

- Evaluate antibody detection tests to support disease surveillance in an outbreak.

National capacity building for African swine fever testing

- Establish African swine fever testing capability across Australia.

National capacity building for lumpy skin disease testing

- Establish lumpy skin disease testing capability across Australia.

Improved detection of Johne's disease

- Develop a novel diagnostic test that uses host microRNAs to detect Johne's disease in cattle.

Testing for lumpy skin disease in tissue samples

- Develop a test to detect lumpy skin disease virus in a tissue sample.

Equine piroplasmosis diagnostic capability development

- Develop antibody detection tests to screen horses entering or exiting Australia.

Assessing national bluetongue diagnostic capability

- Evaluate antibody detection tests to support disease surveillance and response activities.

Improving flavivirus detection in livestock

- Develop an antibody detection test that can screen for several viruses at once, such as Japanese encephalitis virus, Murray valley encephalitis virus and West Nile virus.

Validation of Johne's disease tests

- Develop molecular assays using host microRNA biomarkers to improve detection of Johne's disease.

Differentiating lumpy skin disease infected and vaccinated animals

- Assess existing tests that determine if an animal was infected or vaccinated.

Completed Projects

Point-of-care test platform for emergency animal disease diagnosis

- Develop a portable platform device containing a suite of tests that detect major emergency animal diseases.

Review of the strategies and guidelines of point-of-care testing for infectious diseases in animals

- Inform national point-of-care testing policy for notifiable animal diseases in Australia.

No current DAFF supported projects.

Download project updates

The project updates provide an overview of the progress of each project.

November 2025

- 2025 project updates National Animal Health Diagnostics Business Plan 2021-2026 (PDF 354 KB)
- 2025 project updates National Animal Health Diagnostics Business Plan 2021-2026 (DOCX 153 KB)

November 2024

- 2024 project updates National Animal Health Diagnostics Business Plan 2021-2026 (PDF 333 KB)
- 2024 project updates National Animal Health Diagnostics Business Plan 2021-2026 (DOCX 149)

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Project Proposals

Do you have a project that supports the objectives of the plan?

Please email us to discuss your proposal at animalhealthlaboratories@aff.gov.au.